Where's the Data?

Happy International

LOVE DATA WEEK

Thank you for celebrating with us!

Join us for more events all week!

SCHEDULE OF EVENTS

Monday, February 9
12:00pm - 1:00pm | Online
"Dataverse Reloaded: Improving Data Sharing Through Design"
With Ceilyn Boyd and Sonia Barbosa

2:00pm - 3:00pm | Online
"From Text to 3D: Word Clouds for Exploration"
With Matt Cook, Digital Scholarship Program Manager

Tuesday, February 10
12:00pm - 1:00pm | Online
"Introduction to Text Analysis with Voyant Tools"
With Hugh Truslow, Head of Social Sciences & Visualization

2:00pm - 3:00pm | Countway Library
"Finding GIS Data for Your Project"
With Belle Lipton, GIS Librarian, Harvard Map Collection

Wednesday, February 11
12:00pm - 1:00pm | Online
"Explore the Harvard Geospatial Library"
With Belle Lipton, Marc McGee, and Anna He

2:00pm - 5:00pm | Countway Library
"Community Data Preservation: A Climate & Health Datathon"
With Public Environmental Data Partners

Thursday, February 12
12:00pm - 2:00pm | Lamont Library
"Love Data Rodeo: A Roundup of Useful Ways to Wrangle Your Data"
With Harvard Data Specialists

HARVARD
LIBRARY

Dataverse Reloaded

Improving 🤝 Data Sharing Through Design

🩷 Love Data Week, 9 February 2026

Sonia Barbosa & Ceilyn Boyd

About the Presenters





Sonia Barbosa

Associate Director of Dataverse Support, Data Curation, and the Murray Archive

Institute for Quantitative Social Sciences (IQSS), Harvard University

Ceilyn Boyd

Interim Director of Data Science & Product Research

Institute for Quantitative Social Sciences (IQSS), Harvard University

Agenda

- About the Dataverse Project
- Harvard Dataverse Features & Services, Updates!
- Redesigning the Dataverse UI

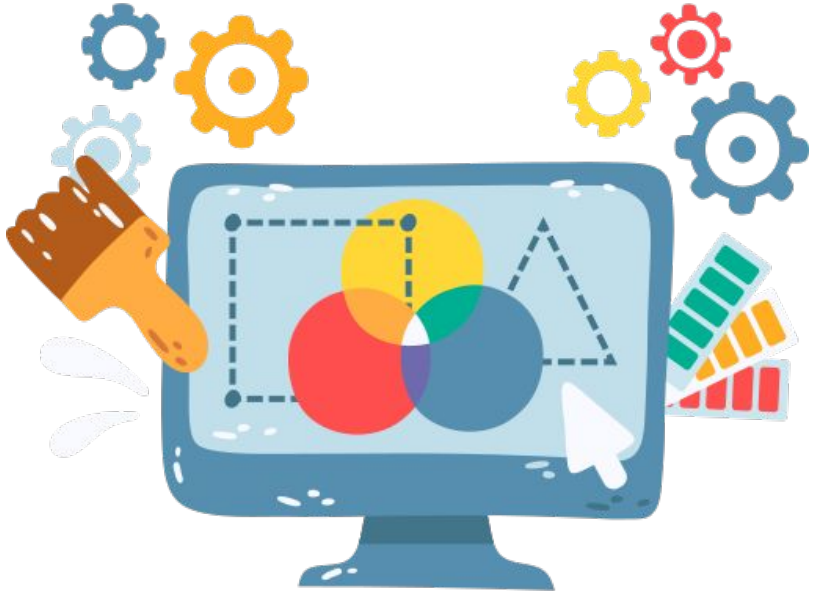


Image: Flaticon

Dataverse Project

- Open Source
- **Data Repository Software Platform**
- 145 Installations
- Active Community
- FAIR support
- Standards support
- Many features
- Previewers
- External tools
- Integrations



Dataverse Project Timeline

Dr. Gary King
Founder & Principal Investigator
Albert J. Weatherhead III University Professor
Harvard University
Institute for Quantitative Social Sciences



2025 Community Meeting, UNC

1997 ➡ 2006 ➡ 2007 ➡ 2008 ➡ 2015 ➡ 2018 ➡ 2023 ➡ 2024 ➡ 2025 ➡ 2026

1997
Project Begins

Focus on preserving & sharing Social Sciences data

2006
The Dataverse Network 1.0

🎉 Harvard Dataverse Repository established

2007
🎉 Announcing the Dataverse Network

2008
Odum Archive: First Dataverse outside Harvard

2015
Data Citations

🤝 First Open Dataverse Community Meeting

Dataverse 4.0

2018
📣 GDCC established

2023
🎉 100th Dataverse installation

Dataverse 6.0

Dataverse Re-architecture

2024
🤖 Ask the Data

🥐 Croissant

New Dataset Types (e.g., software, computational workflows)

2025
137 Dataverse Installations

Dataverse 6.7

Large Data Services

Traditional Knowledge (TK) Labels

Dataverse MCP Server

Single-Page App Launch

2026
🚀 The Next Frontier

Locally FAIR Data

Trusted Data Labels

📣 145 Dataverse Installations

🌍 Geospatial improvements

Dataverse Repository Platform Features

- Support for FAIR Data Principles
- Data citation for datasets and files
- OAI-PMH (Harvesting)
- APIs for interoperability and custom integrations
- API client libraries
- DataCite integration
- Login via Shibboleth
- Login via ORCID, Google, GitHub, or Microsoft
- Login via OpenID Connect (OIDC)
- Internationalization
- Versioning
- Restricted files
- Embargo
- Custom licenses
- Custom terms of use
- Publishing workflow support

- File hierarchy
- File previews
- Preview and analysis of tabular files
- Usage statistics and metrics
- Guestbook
- Fixity checks for files
- File download in R and TSV format
- Faceted search
- Customization of collections
- Private URL
- Widgets
- Notifications
- Schema.org JSON-LD
- External tools
- External vocabulary
- Dropbox integration
- GitHub integration

- Integration with Jupyter notebooks
- User management
- Curation status labels
- Branding
- Backend storage on S3 or Swift
- Direct upload and download for S3
- Export data in BagIt format
- Post-publication automation (workflows)
- Pull header metadata from Astronomy (FITS) files
- Provenance
- Support for rsync
- Auxiliary files for data files
- Metadata export (multiple formats)

https://dataverse.org/software-features

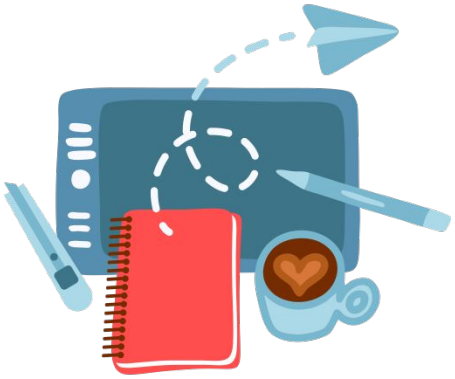


Image: Flaticon

Harvard Dataverse

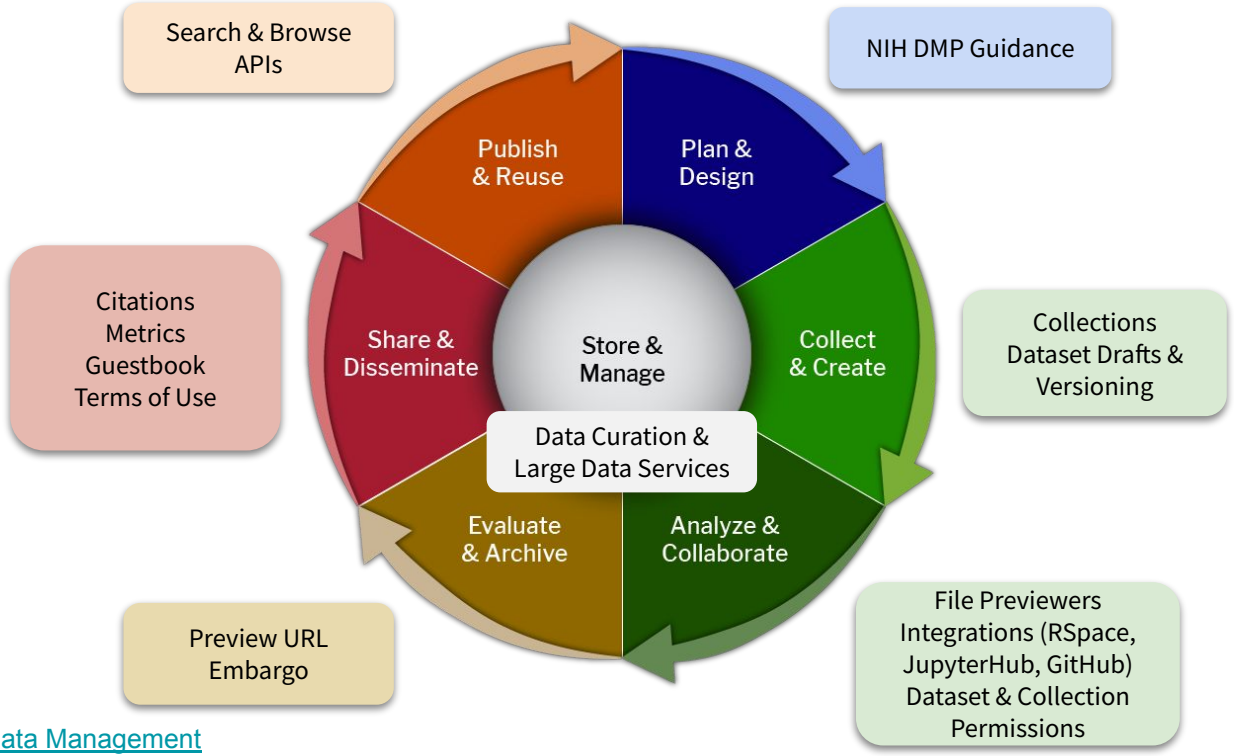
- About the repository
- Harvard user benefits
- New features

Harvard Dataverse throughout the research lifecycle



Credit: Harvard Biomedical Data Management

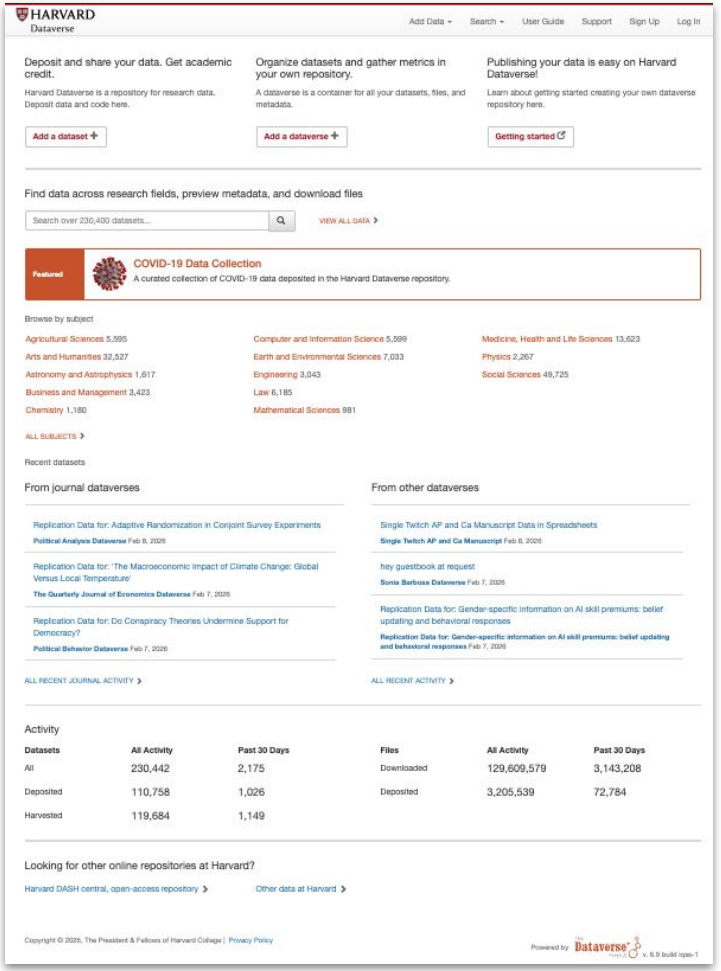
Harvard Dataverse features and services



Credit: [Harvard Biomedical Data Management]

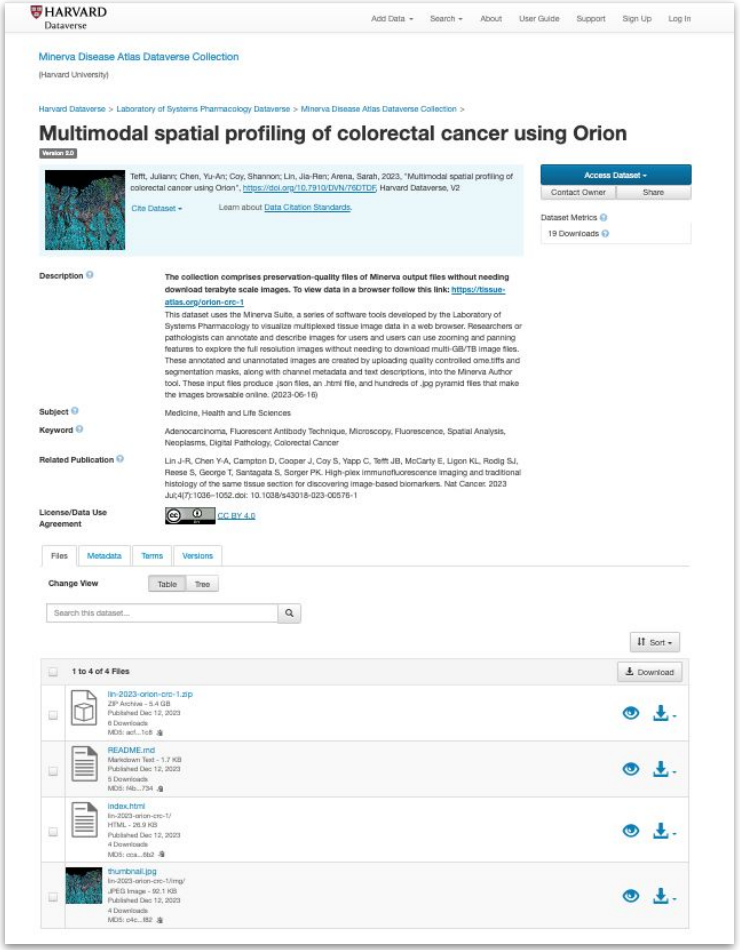
Harvard Dataverse

- Harvard's institutional research data repository
- Largest Dataverse installation
- Established in 2006
- Access to over 230,000 datasets



Harvard Dataverse

- Managed data repository open to all researchers from any discipline, both inside and outside of the Harvard community, where you can share, archive, cite, access, and explore research data. Part of the NIH-GREI coopetition.
- Collaboration between Harvard Library, Harvard University Information Technology, and IQSS
- All file types and disciplines are welcome
- We do not currently support sensitive data. All data must be de-identified and utilize managed access per best practices.



Harvard Dataverse storage

Harvard Users	Other Users	Large Data
2.5T per account free	1T per account free	Fee-based tape & disk options

Maximum File Size

Free Storage Options	*Large Data*
Self-deposit: 2.5GB per file Contact us for larger files	Varies according to storage option

Getting started

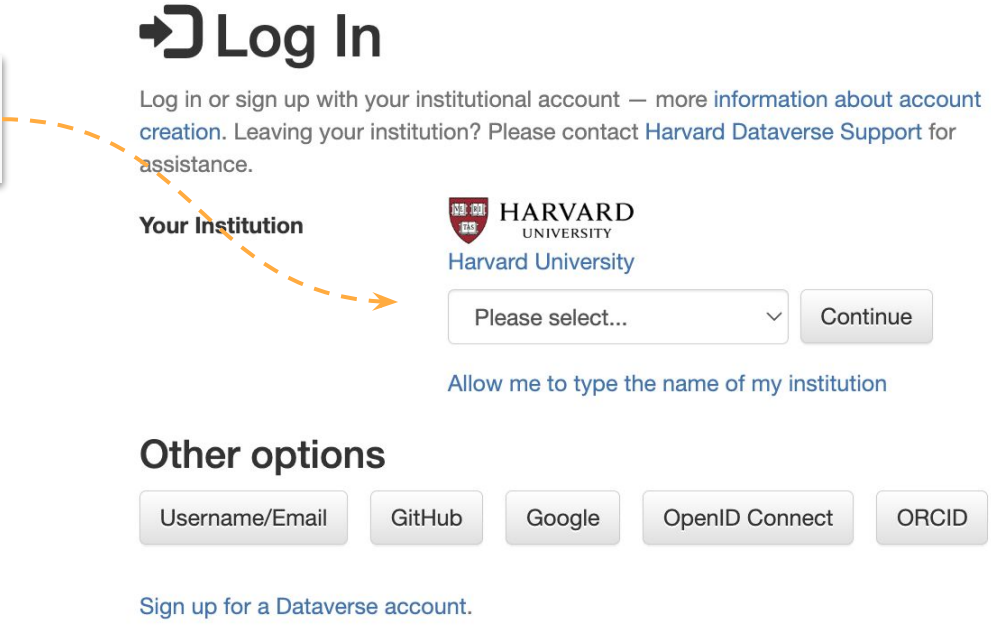
Five steps to success

1. Create account	2. Create collection	3. Create dataset	4. Publish dataset	5. Create dataset & file versions*
• Harvard Key • GitHub • ORCID • Gmail	• Customization • Subcollections • + Custom metadata options • + Storage options • Guestbook • Deposit Templates	• Upload files • Add metadata • Add licenses • Add links to other publications	• Receive data citation • Metadata shared automatically with search engines & other repositories	• While your project continues: ○ Add metadata ○ Add files

dataverse.harvard.edu

Create an account

User Your Harvard Key +



Create collection

Collections

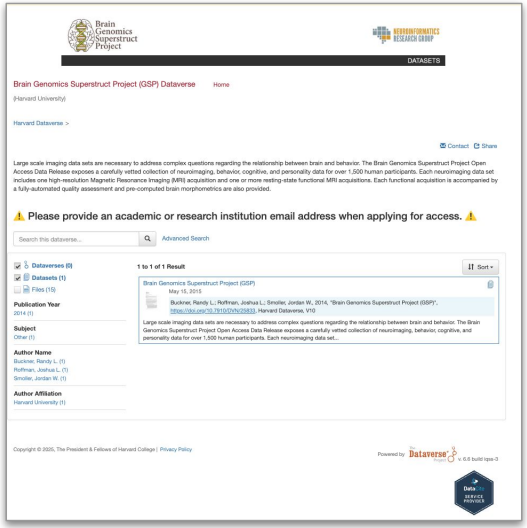
- Own administration
- Own branding (and can be embedded anywhere)

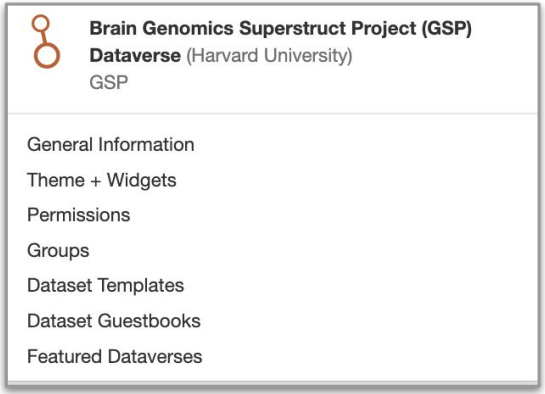
Datasets

- Citation
- Metadata
- Versioning
- Private URL
- Custom Terms/Multiple License/Permissions
- Guestbooks
- Publishing Workflows

Files

- Citation
- Ingest
- Preview/Explore
- Metadata
- Versioning
- Permissions/Embargo/Restrictions





Supported Metadata

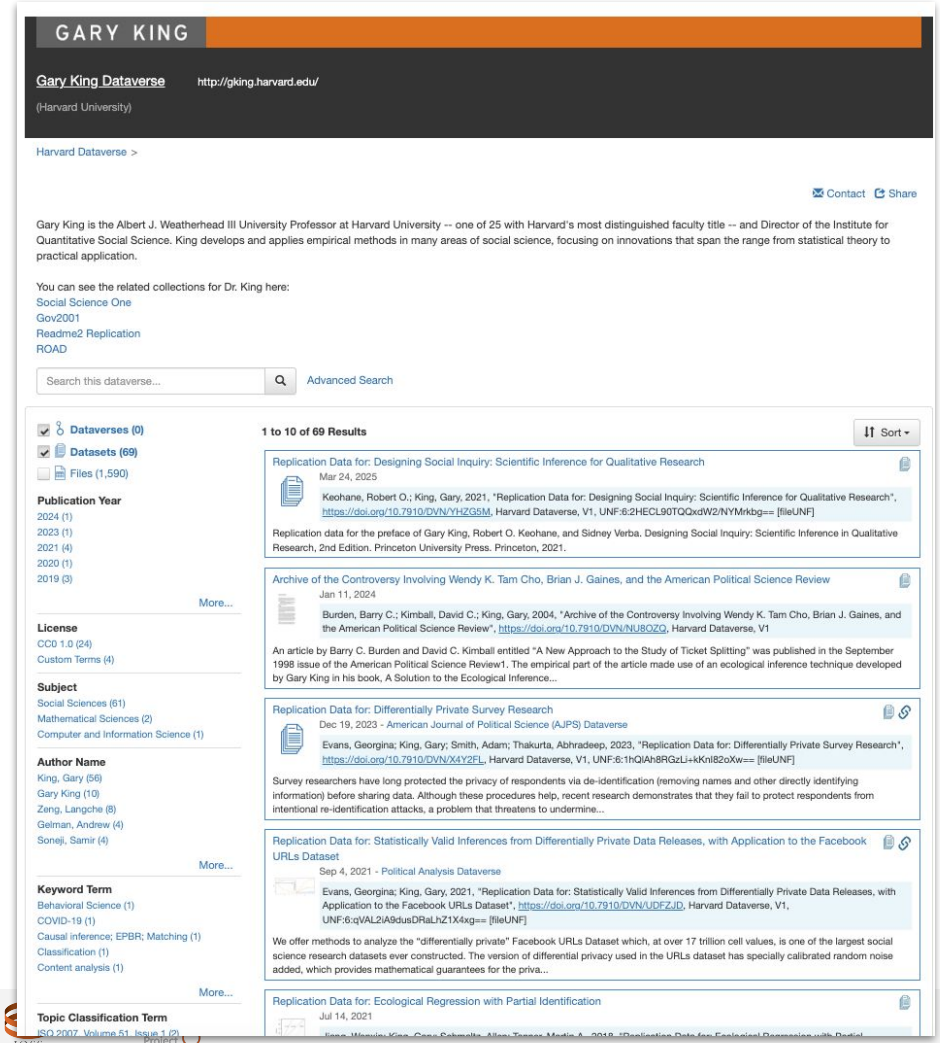
Detailed below are what metadata schemas we support for Citation and Domain Specific Metadata in the Dataverse Project:

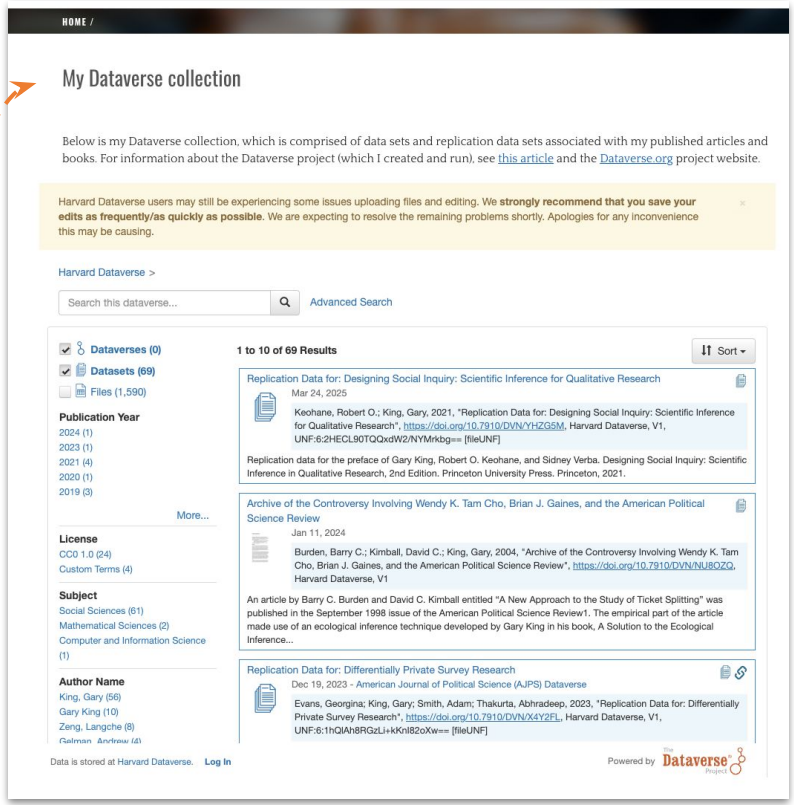
- Citation Metadata (see .tsv version): compliant with DDI Lite, DDI 2.5 Codebook, DataCite 3.1, and Dublin Core's DCMI Metadata Terms . Language field uses ISO 639-1 controlled vocabulary.
- Geospatial Metadata (see .tsv version): compliant with DDI Lite, DDI 2.5 Codebook, DataCite, and Dublin Core. Country / Nation field uses ISO 3166-1 controlled vocabulary.
- Social Science & Humanities Metadata (see .tsv version): compliant with DDI Lite, DDI 2.5 Codebook, and Dublin Core.
- Astronomy and Astrophysics Metadata (see .tsv version): These metadata elements can be mapped/exported to the International Virtual Observatory Alliance's (IVOA) VOResource Schema format and is based on Virtual Observatory (VO) Discovery and Provenance Metadata.
- Life Sciences Metadata (see .tsv version): based on ISA-Tab Specification, along with controlled vocabulary from subsets of the OBI Ontology and the NCBI Taxonomy for Organisms.
- Journal Metadata (see .tsv version): based on the Journal Archiving and Interchange Tag Set, version 1.2.

Metadata Fields

Choose the metadata fields to use in dataset templates and when adding a dataset to this dataverse.

- [x] Citation Metadata (Required) [+] View fields + set as hidden, required, or optional
- [ ] Geospatial Metadata [+] View fields
- [ ] Social Science and Humanities Metadata [+] View fields
- [ ] Astronomy and Astrophysics Metadata [+] View fields
- [ ] Life Sciences Metadata [+] View fields
- [ ] Journal Metadata [+] View fields
- [ ] MRA Metadata [+] View fields
- [ ] Graduate School of Design Metadata [+] View fields
- [ ] Alliance for Research on Corporate Sustainability Metadata [+] View fields
- [ ] Political Science Replication Initiative Metadata [+] View fields
- [ ] PSI Metadata [+] View fields
- [ ] CHIA Metadata [+] View fields
- [ ] Digaai Metadata [+] View fields
- [ ] Computational Workflow Metadata [+] View fields
- [ ] Local Contexts [+] View fields
- [ ] 3D Objects Metadata [+] View fields





Imbed your collection widget into your faculty or project website to enhance discoverability of your datasets

Sonia Barbosa & Ceilyn Boyd

Create dataset

Collections

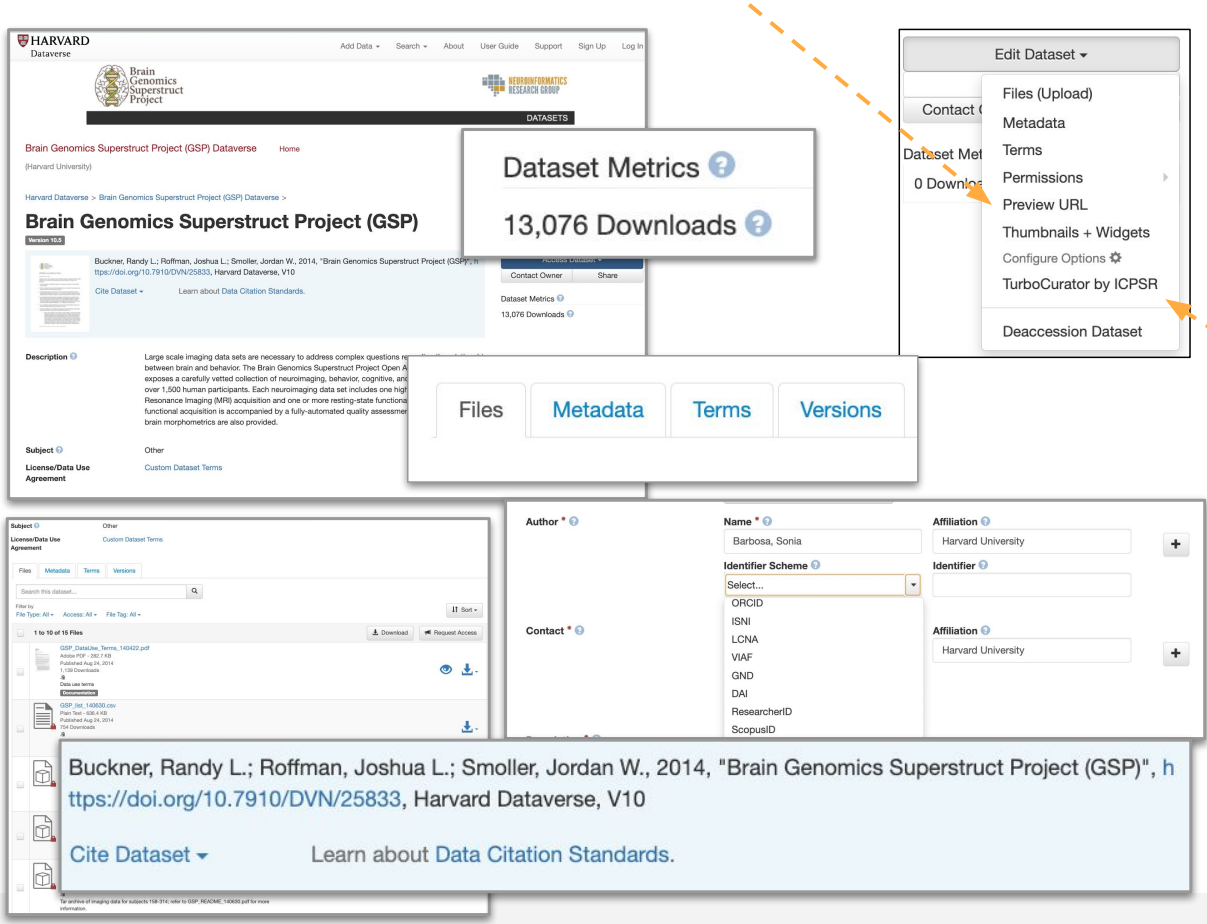
- Own administration
- Own branding (and can be embedded anywhere)

Datasets

- Citation
- Metadata
- Versioning
- Private URL
- Custom Terms/Multiple License/Permissions
- Guestbooks
- Publishing Workflows

Files

- Citation
- Ingest
- Preview/Explore
- Metadata
- Versioning
- Permissions/Embargo/Restrictions



Sonia Barbosa & Ceilyn Boyd

Files

Collections

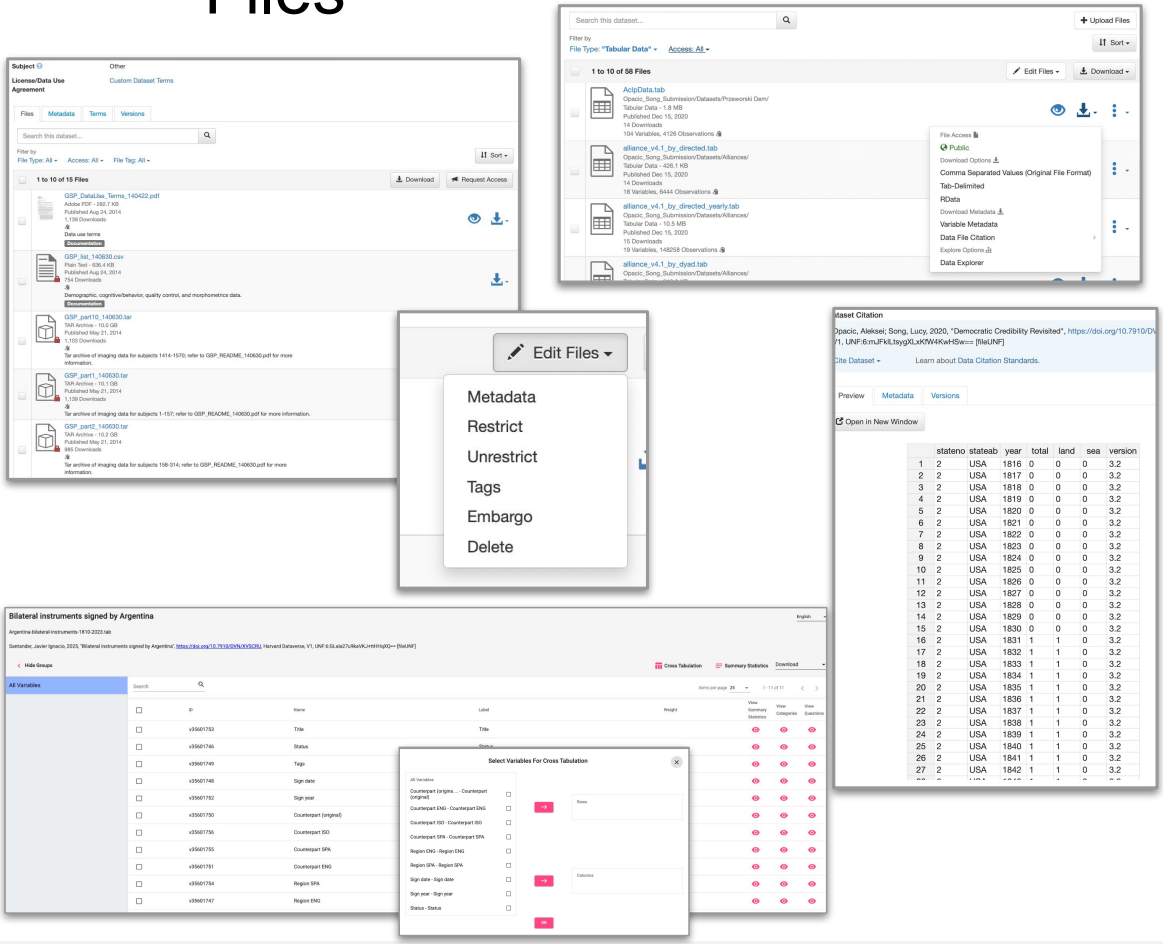
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Datasets

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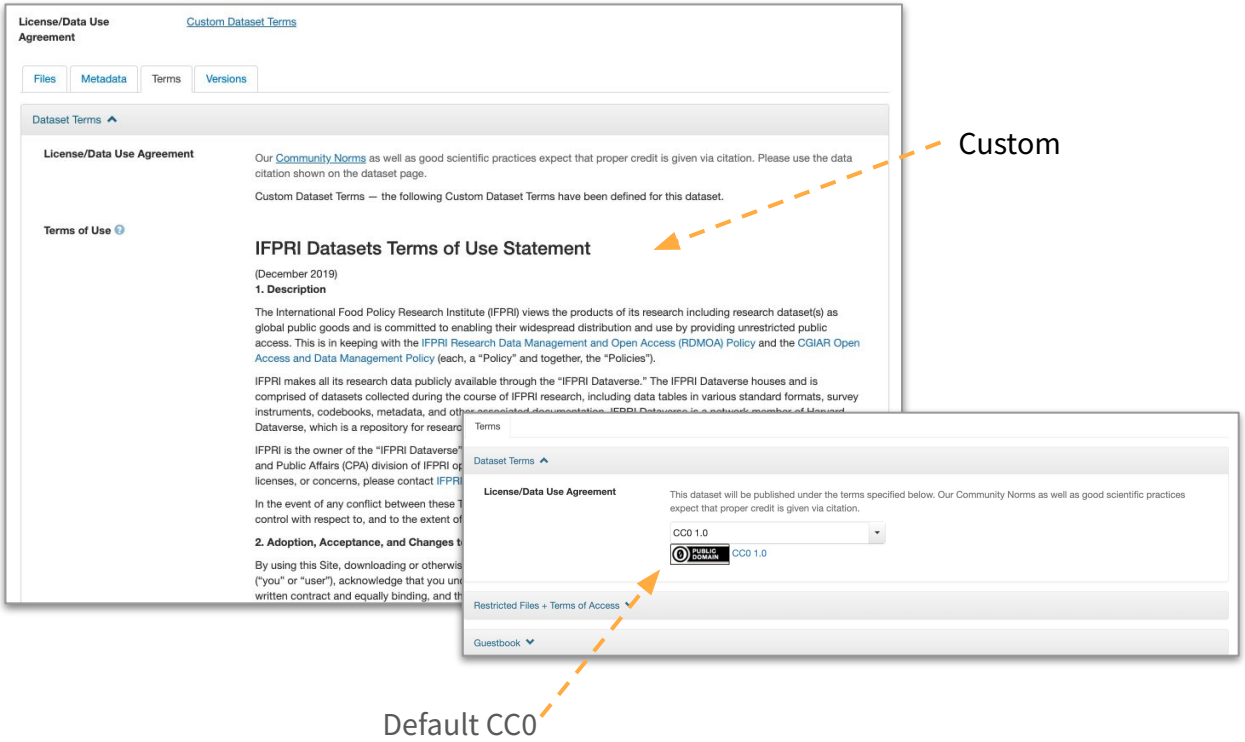
Files

- Citation
- Ingest
- Preview/Explore
- Metadata
- Versioning
- Permissions/Embargo/Restrictions



Sonia Barbosa & Ceilyn Boyd

Apply license/s, set terms of use

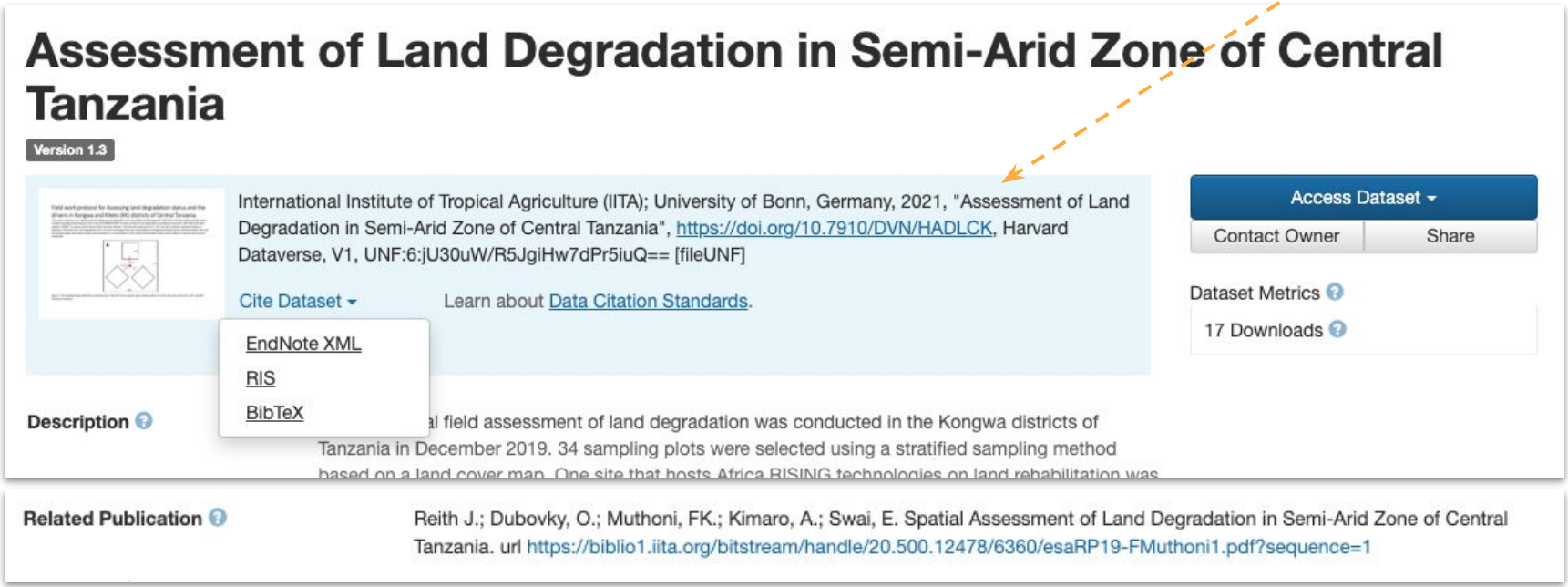


Choose simple or custom terms of use

- CC0 1.0
- CC BY 4.0
- CC BY-NC 4.0
- CC BY-NC-ND 4.0
- CC BY-NC-SA 4.0
- CC BY-ND 4.0
- CC BY-SA 4.0
- PDDL-1.0
- ODC-By 1.0
- ODbL 1.0
- OGL UK 3.0
- MIT
- Custom Dataset Terms

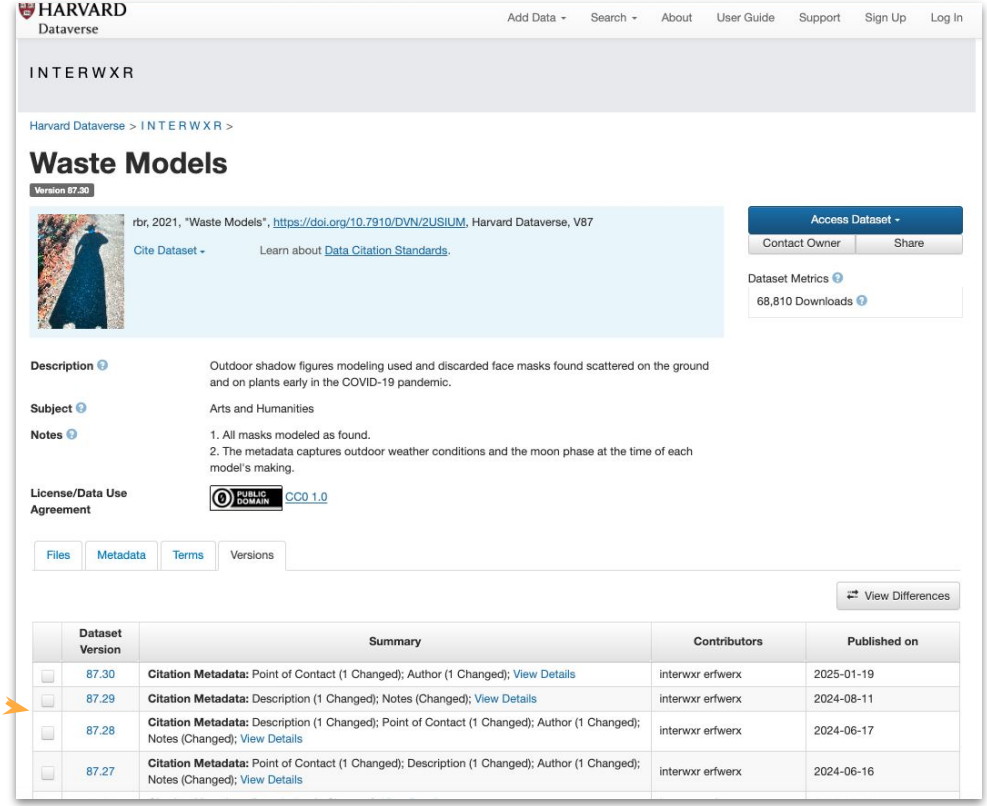
Publish your dataset, receive a citation

Digital Object Identifier (DOI)



Create and share versions of your data

- Minor version: metadata updates
- Major version: new files, replaced files, etc.



Datasets - Ask the Data

Tell me what you want to know

What is this data about?

Answer please

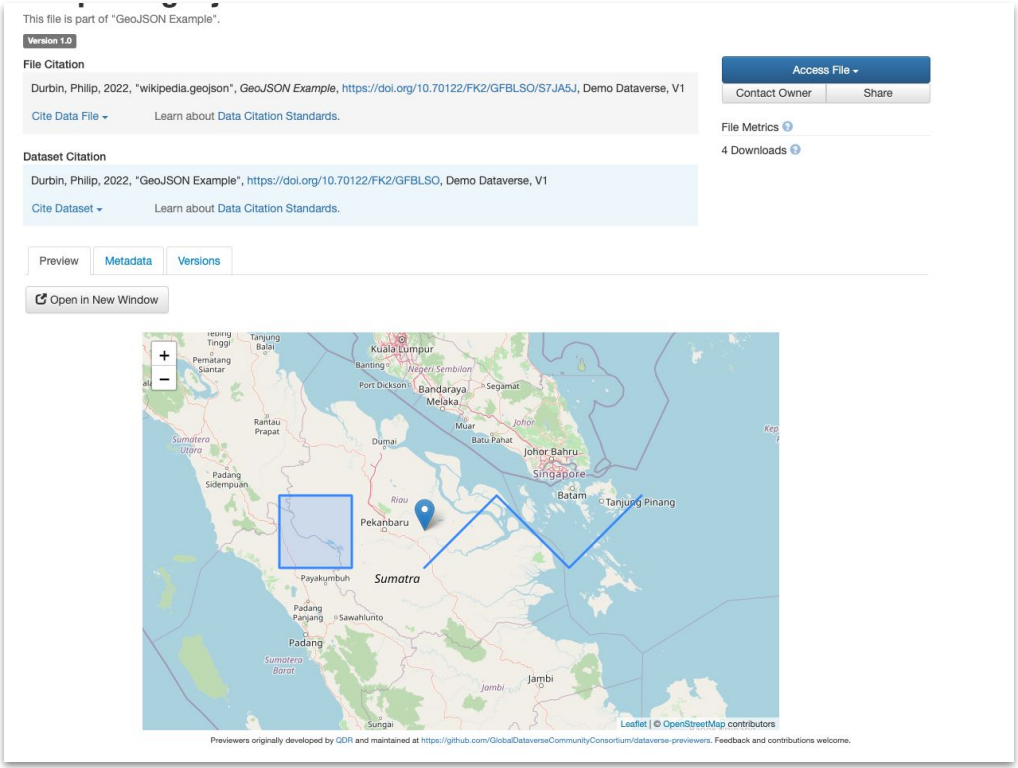
This data is about traffic accidents in the US.

STATE	COUNTY	MONTH	DAY	HOUR	MINUTE	VE_FORMS	PERSONS	PEDS	NHS	ROAD_FNC	ROUTE	SP_JUR	HARM_EV	MAN_COLL	REL_JUNC
1	97	1	26	15	30	2	2	0	0	6	4	0	12	2	1
1	101	2	1	23	55	1	1	0	1	2	2	0	42	0	1
1	81	2	2	5	10	1	1	0	0	6	4	0	42	0	1
1	101	1	18	22	14	1	2	1	1	12	2	0	8	0	1
1	101	1	20	13	5	1	1	0	1	11	1	0	34	0	1
1	73	1	31	2	40	2	3	0	1	11	1	0	25	0	1
1	73	1	14	19	0	1	2	0	1	11	1	0	34	0	1
1	73	1	24	19	45	2	3	1	0	6	6	0	8	0	2
1	97	1	23	19	34	2	3	0	0	13	2	0	12	5	2
1	97	1	31	23	33	1	2	0	0	4	3	0	30	0	1

Viewing rows 1 through 10 of 38444

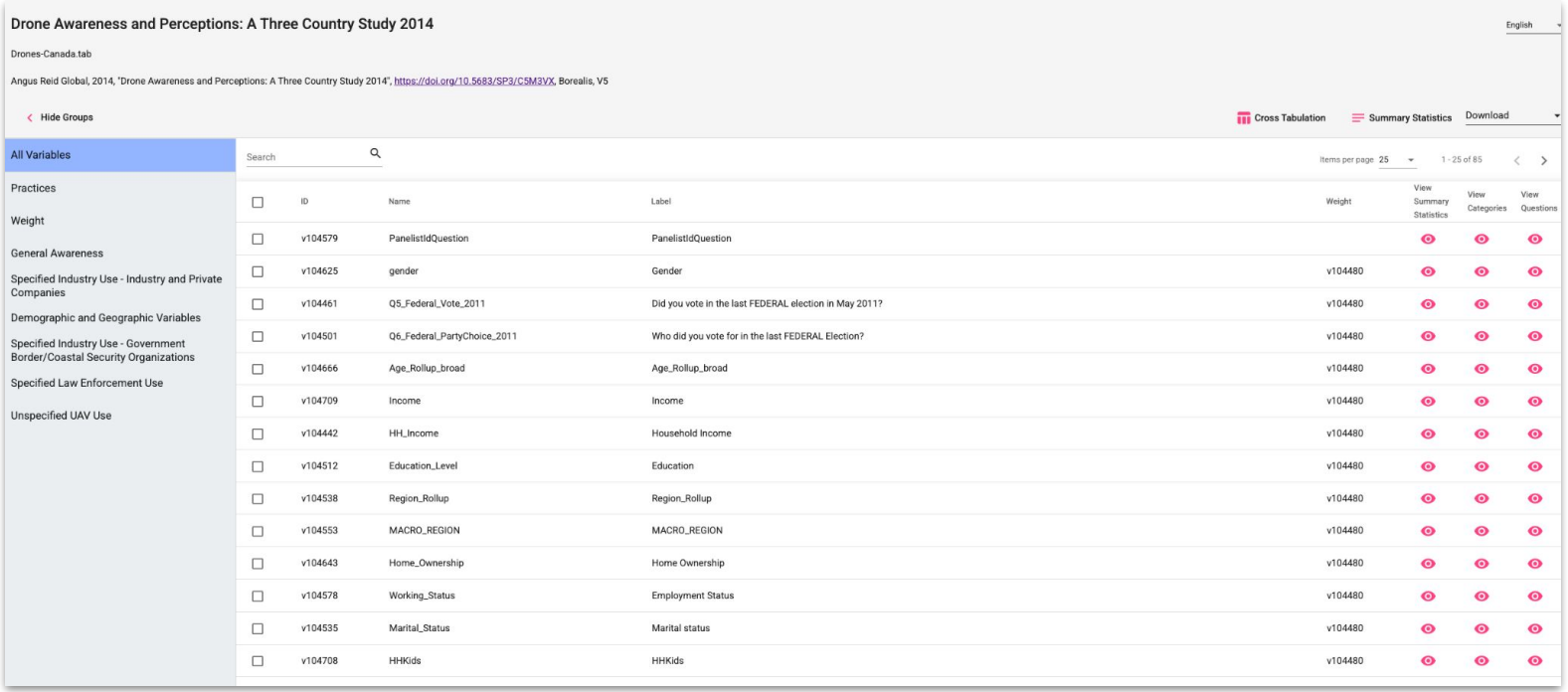
https://github.com/IQSS/askdataverse/tree/main/askthedata

Datasets - GeoJSON Previewer

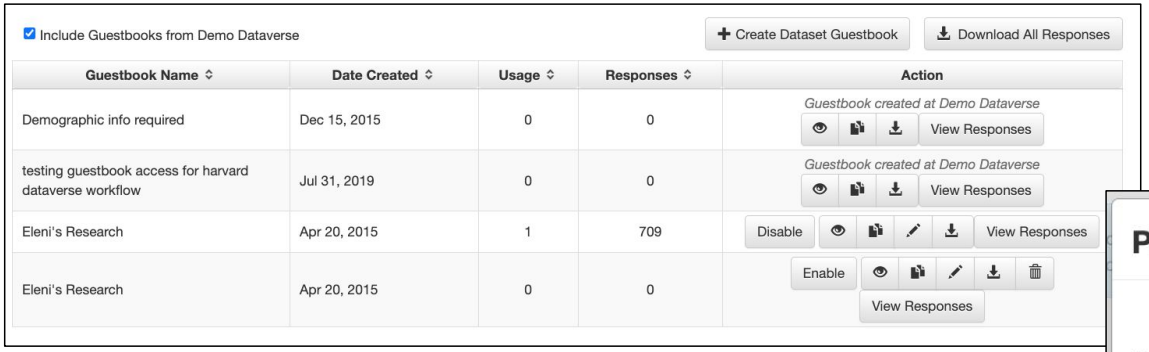


https://github.com/GlobalDataverseCommunityConsortium/dataverse-previewers

Datasets - Tabular Files/Data Explorer

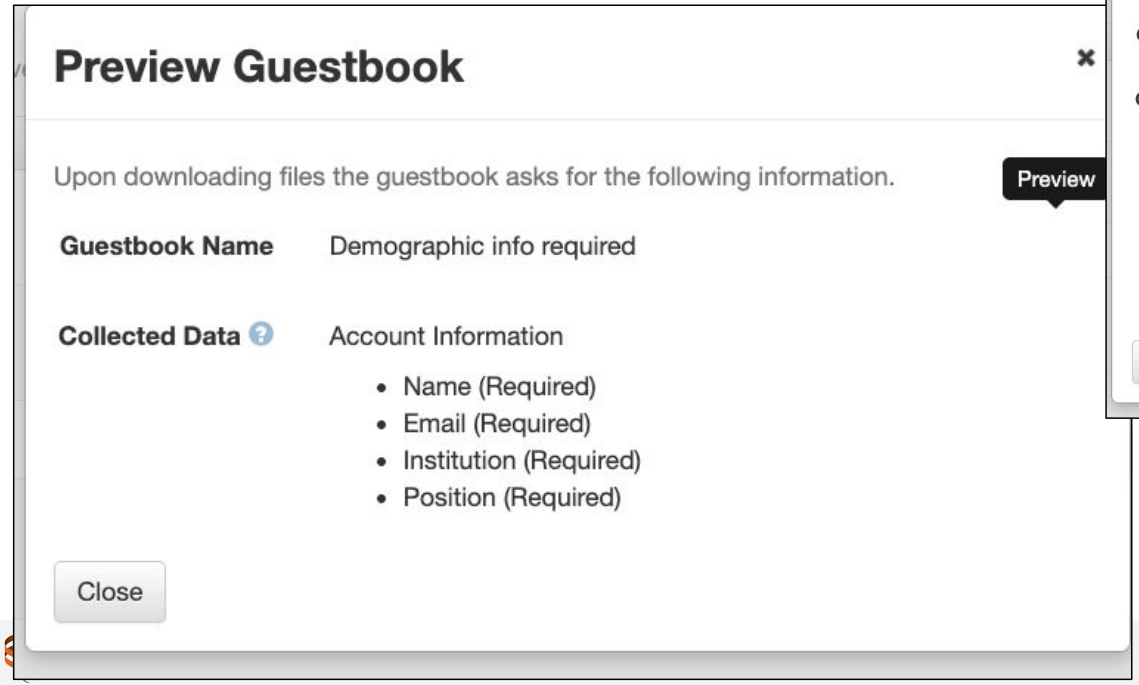


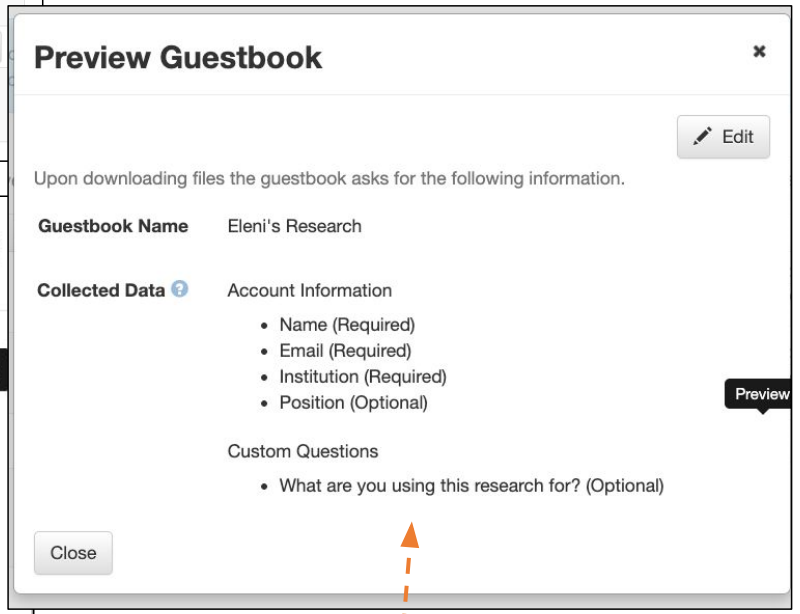
https://guides.dataverse.org/en/latest/user/tabulardataingest/index.html
https://github.com/scholarsportal/dataverse-data-explorer-v2



Create as many as needed and apply at the dataset level

HARVARD Dataverse





Custom questions

Previewers, integrations & roles

Previewers & External Tools

- Tabular files (e.g., STATA, Excel)
- Geospatial files (e.g., NetCDF)
- 3D viewers
- Preview Zip files
- View images, maps, videos
- TurboCurator
- Ask the Data
- …And more

Details: File Previewers & External Tools

Integrations

- GitHub
- Dropbox
- Open Science Framework (OSF)
- RSpace
- JupyterHub
- Whole Tale
- …And more

List: Integrations

Roles & Permissions

Assign collaborators, groups, and others different roles with permissions

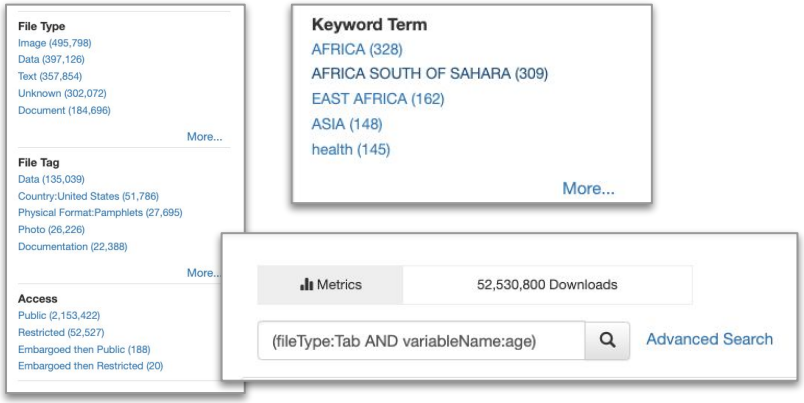
- Admin
- Contributor
- Curator
- File downloader
- Member

Details: Roles & Permissions

Search, browse & APIs

Search & Browse

- Keyword search
- Advanced search
- Funding agency
- Faceted search
- Sorting
- Search datasets from other repositories



APIs

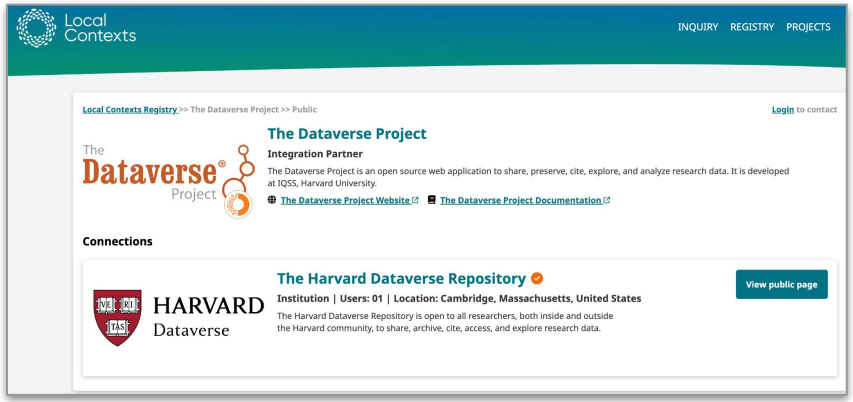
Programmatic access to Harvard Dataverse using various APIs

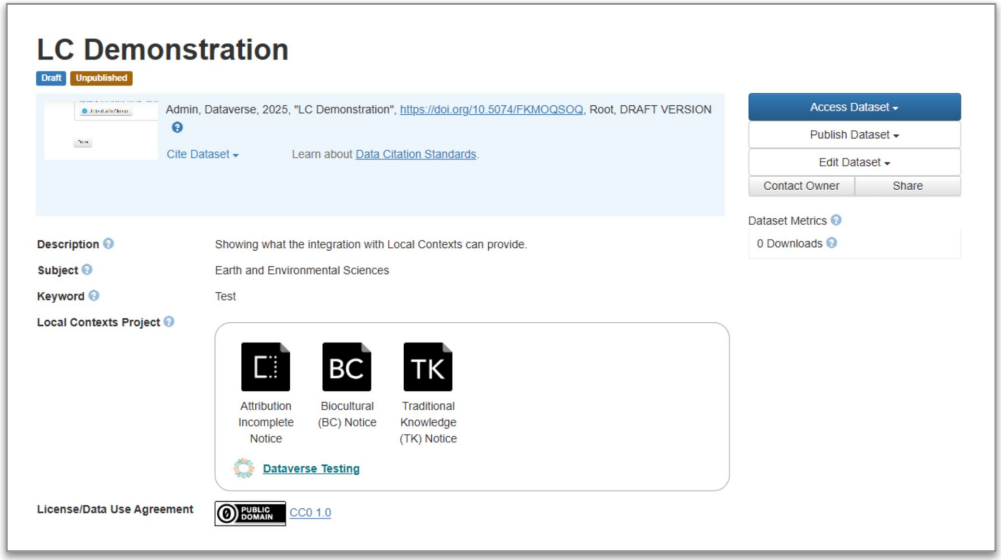
- Native (Ex. create datasets, collections)
- Search (Ex. searching, faceting, sorting)
- Metrics (Ex. downloads, datasets created, files uploaded)
- SWORD (RESTful API to deposit files and metadata: Python, Java, R, Ruby, PHP)
- Data Access (download access to files)
- Python library: pyDataverse

Note: No API access for Large Data resources

Visit the API Guide: Link

Dataverse & Local Contexts Integration





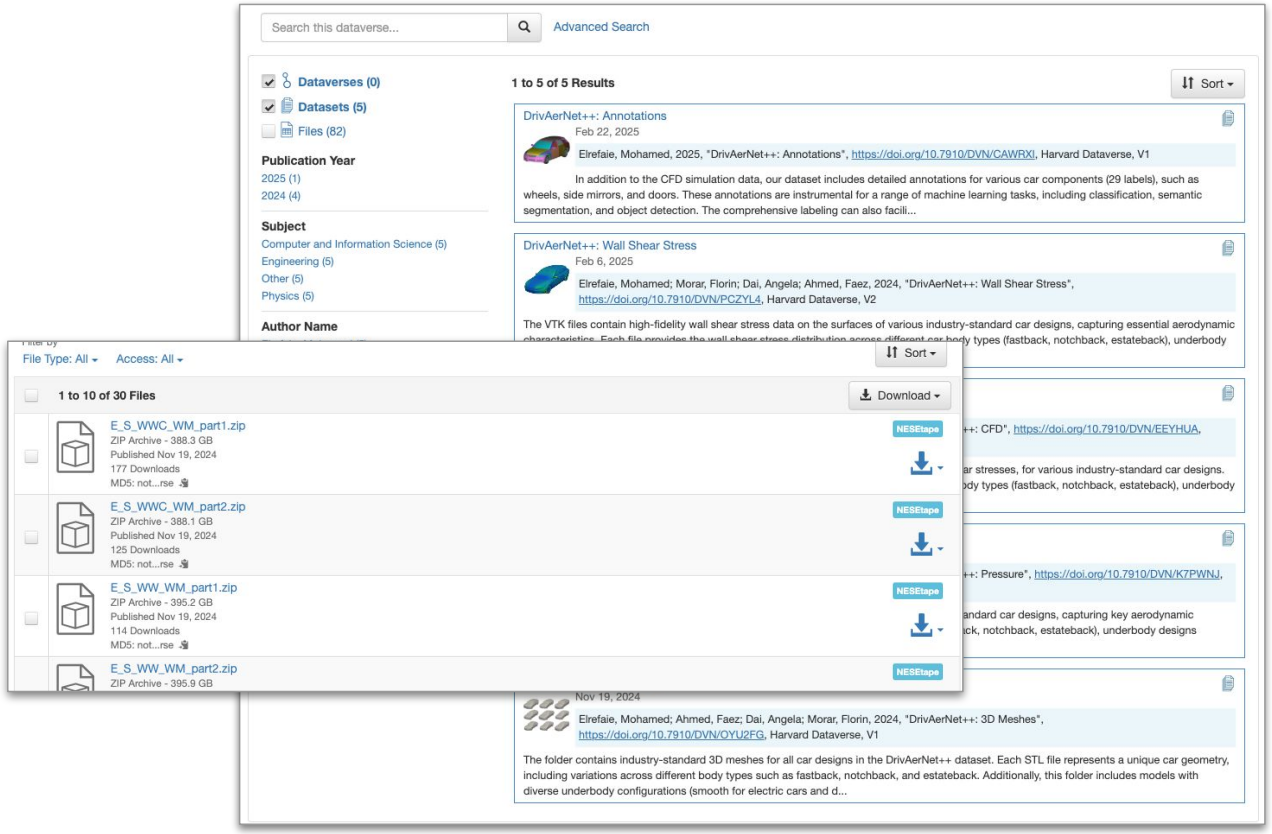
https://guides.dataverse.org/en/latest/installation/localcontexts.html

Large Data Support Service

HARVARD Dataverse

- 12TB of data
- Utilizing NESE tape storage/Globus access
- Guestbook
- No API access for download or upload
- [Upload and Download] instructions

Use Case: [10.5281/zenodo.14782997]



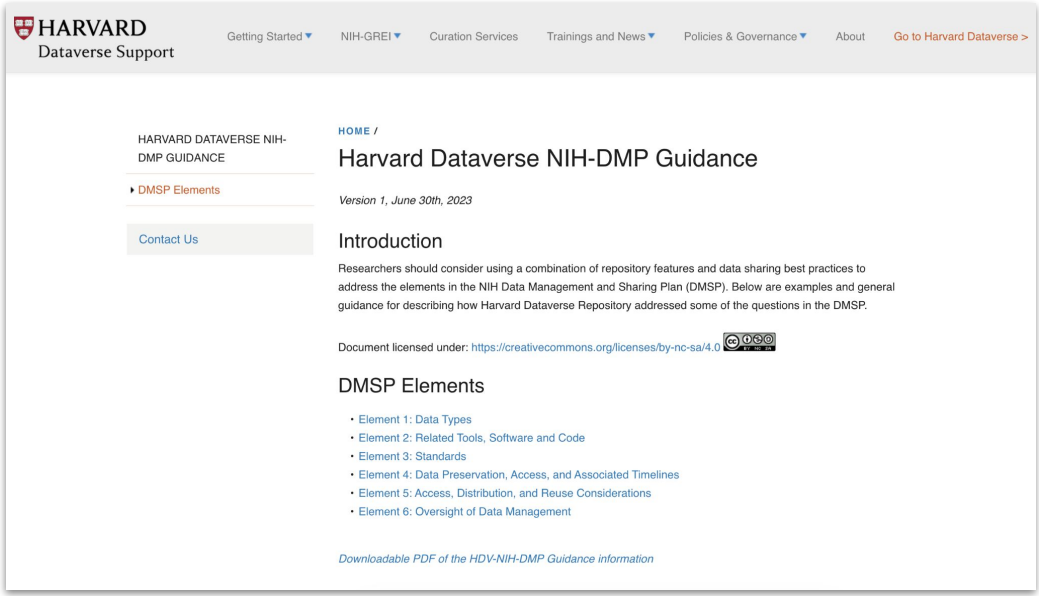
NIH DMP Guidance

NIH Data Management & Sharing Plan

Includes:

- Written plan (2 pgs)
- “maximize the appropriate sharing of scientific data”
- Ideally share in established repositories
- Share as soon as possible

Details: [NIH Scientific Data Sharing Website]



https://support.dataverse.harvard.edu/harvard-dataverse-nih-dmp-guidance

Curation and large data services

Curation Services

Range of curation services, including large data

- Free and extended consultations
- Setup and maintain collections
- Upload and maintain datasets

Service Details: Link

Large Data Services

Support for large datasets (over 2.5T or millions of files)

- NESE Globus Tape
- NESE Globus & S3 Disk
- AWS S3

Minimum 6 years storage

Service Details: Link

Trainings & Office Hours: Calendar
Support: support@dataverse.harvard.edu

Designing for Data Sharing

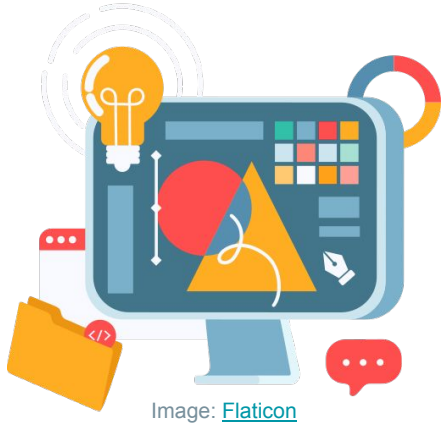
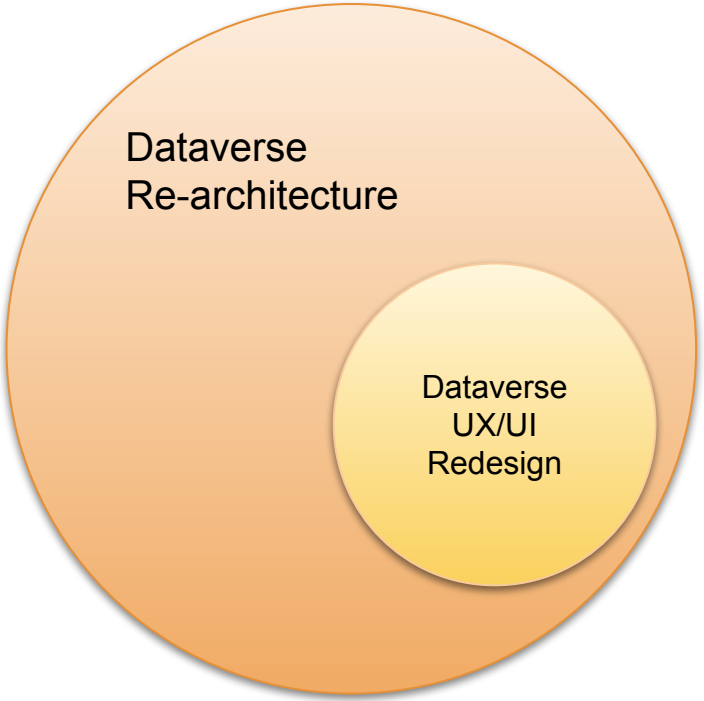


Image: Flaticon

- Improved modularity
- Improved UX - performance, responsiveness & accessibility
- Improved APIs

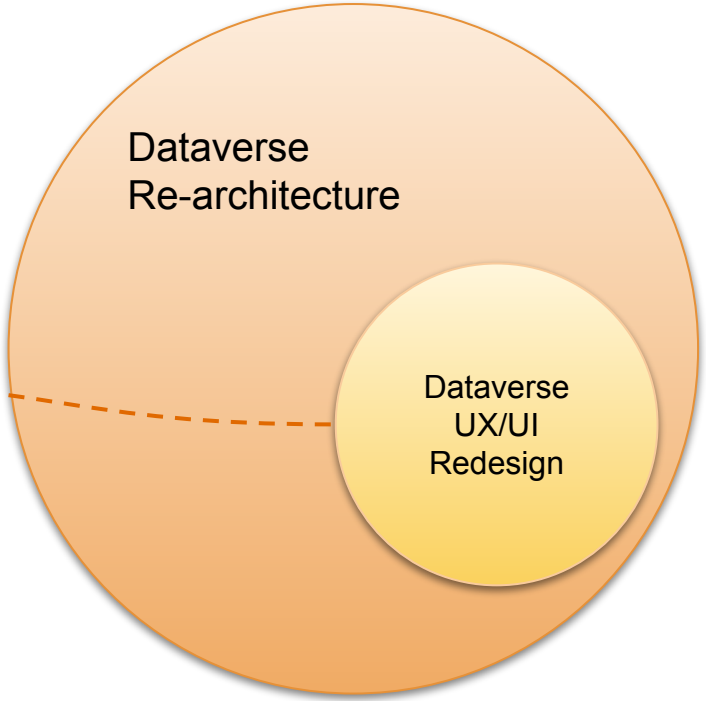
Dataverse Re-architecture

- Modernize the platform
- Separate backend and frontend to increase interoperability & opportunity
- Increase backend and frontend modularity
- Accelerate new UI/UX feature development
- Improve automatic testing (incl. UI)
- Empower the Dataverse community to create new features, UIs, and Dataverse-powered apps



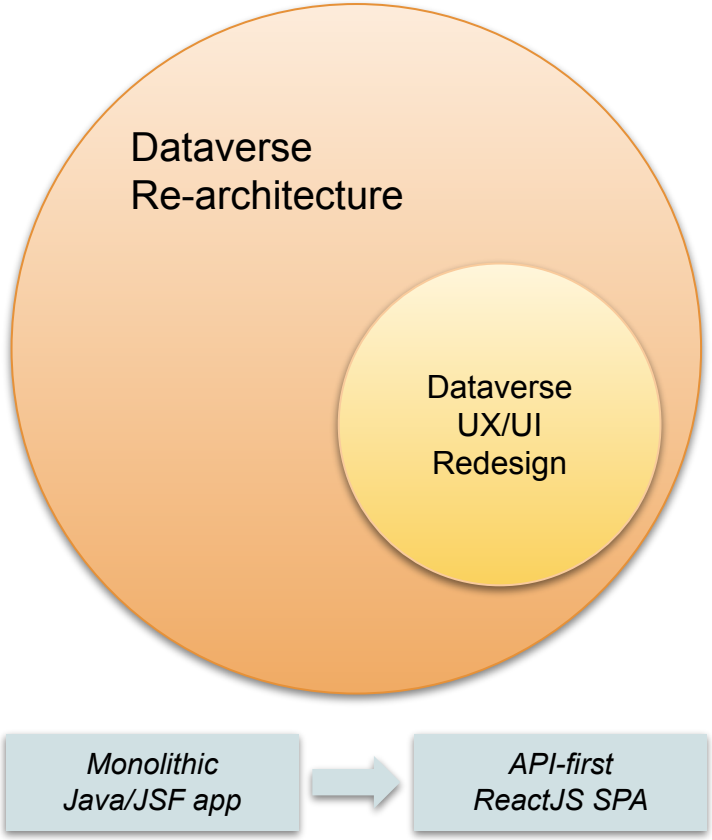
Dataverse Re-architecture

- Modernize the platform
- Separate backend and frontend to increase interoperability & opportunity
- Increase backend and frontend modularity
- Accelerate new UI/UX feature development
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Dataverse UX/UI Redesign

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- Features performance, responsiveness, accessibility, and API improvements

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1. **Performance**: Software should operate quickly & reliably
2. **Interactivity**: Software should provide smooth & meaningful user feedback
3. **Usability**: Software should be easy to learn, efficient & have a clear function
4. **Accessibility**: Software design should inclusive for all users
5. **Structure**: Logical layout and easy user navigation
6. **Visual Design**: Clean, consistent & visually engaging user interface

Performance Improvements

Dataverse APIs

More API endpoints

Uses native queries where possible, resulting in faster searches and page loads

Empowers developers to create new Dataverse-powered apps

Client-Side Rendering

⚡ Lightning-Fast Interactions

- UI updates instantly using React's Virtual DOM

🔄 Smooth Navigation

- Transitions feel native and seamless

📉 Reduced Server Load

- Offloads rendering to the browser
  Server focuses on APIs, not HTML rendering

`React.lazy`

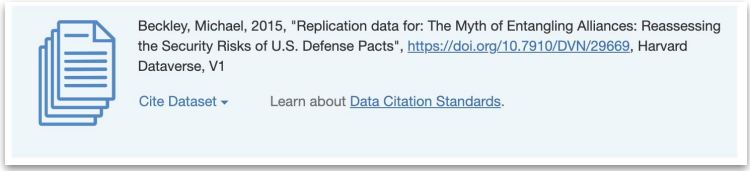
`React.lazy` is a built-in function to load components **dynamically** only when they're needed.

- Reduces initial bundle size
- Speeds up first paint
- Improves Time to Interactive (TTI)

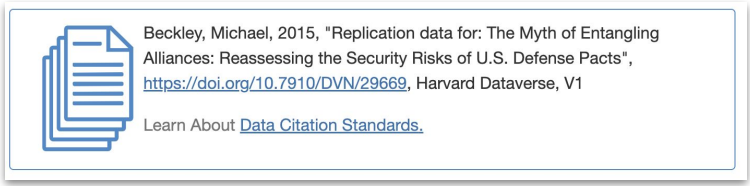
Accessibility Improvements

- All new UI components are accessibility tested with Storybook (storybook.js.org/)
- Lighthouse Compliance Comparison
  - Current UI: 80-90%
  - New UI: 100%

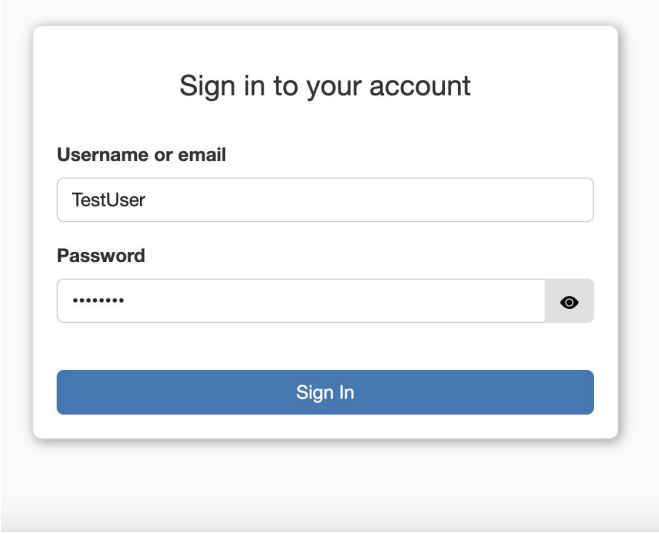
Current UI: poor contrast

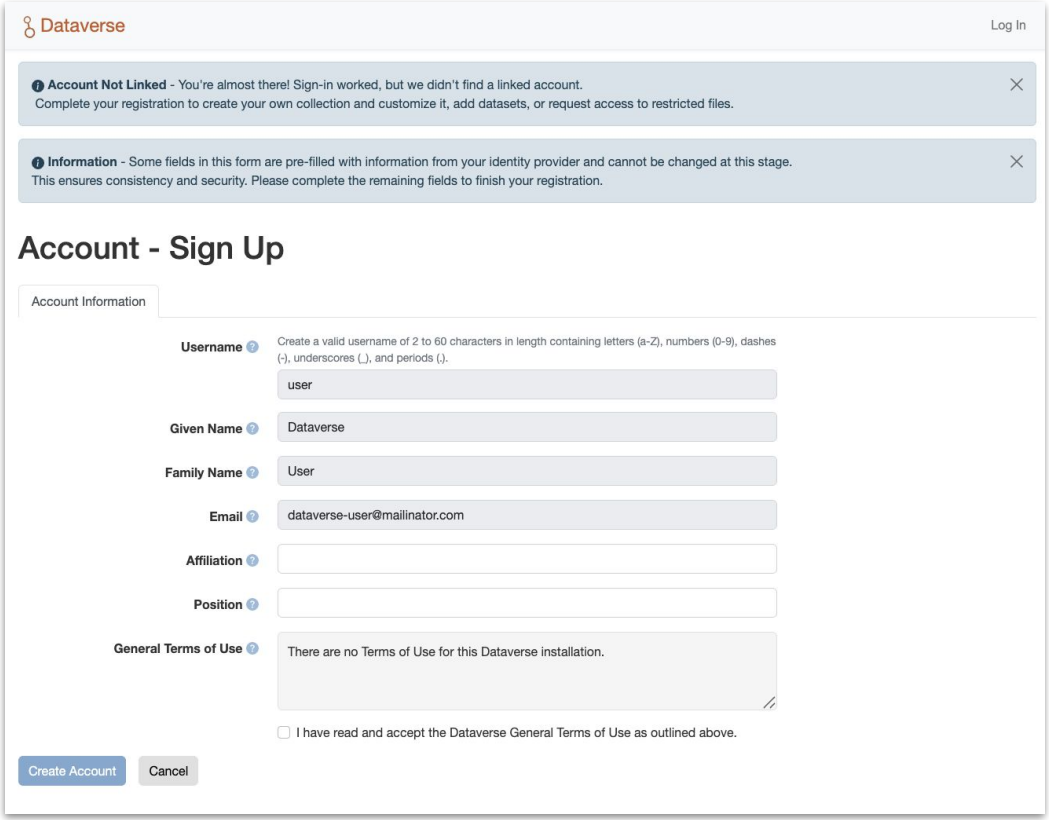


New UI: good contrast



Sign Up & Sign In



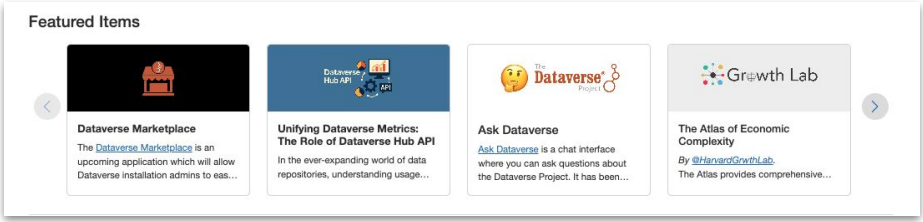


Home Page



Search Bar
supports task-oriented behavior and improves usability. Lets users quickly find specific datasets, studies, or publications.

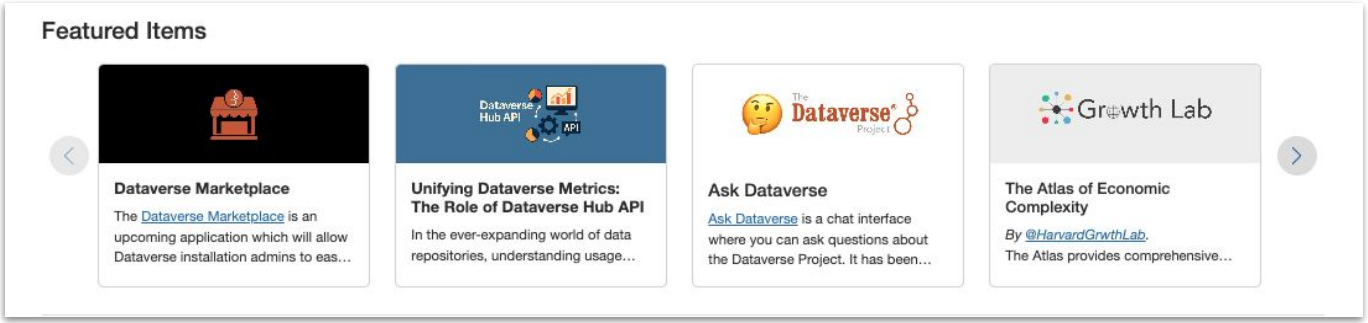
Featured Items
promotes valuable content, engages returning users, and helps new users see what's important or trending.



Dataverse in Numbers *(powered by Dataverse Hub)*
displays stats, builds credibility, shows impact, and helps users understand scale and community involvement.

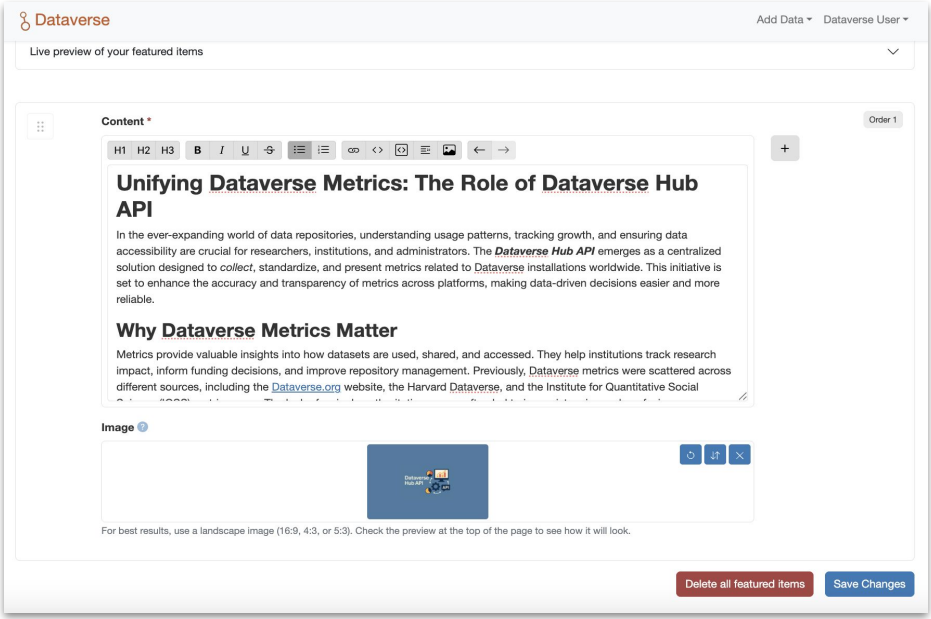


Featured Items



- **Customizable for different needs**: Can be adapted for various user needs with customizable display
- **Enhances discoverability:** Helps surface curated or high-priority content
- Supported for collections, datasets, and files

Featured Items: Example



Rich text and images support

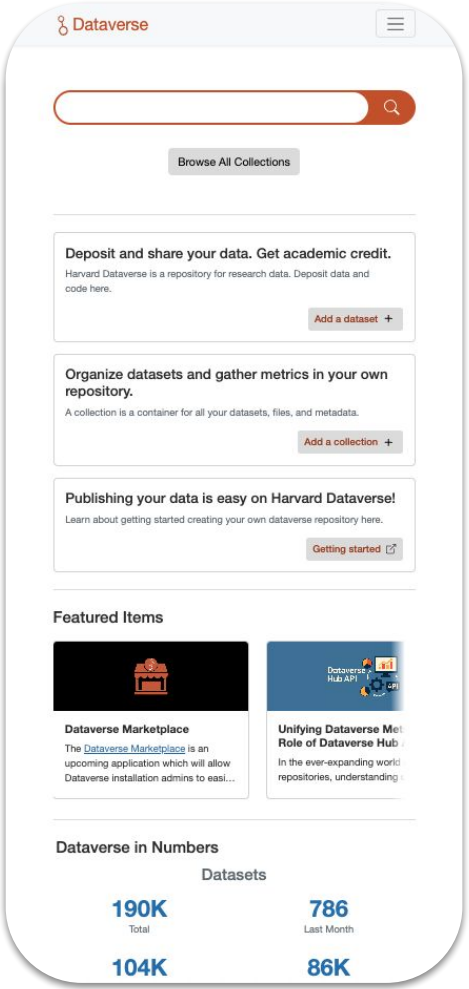


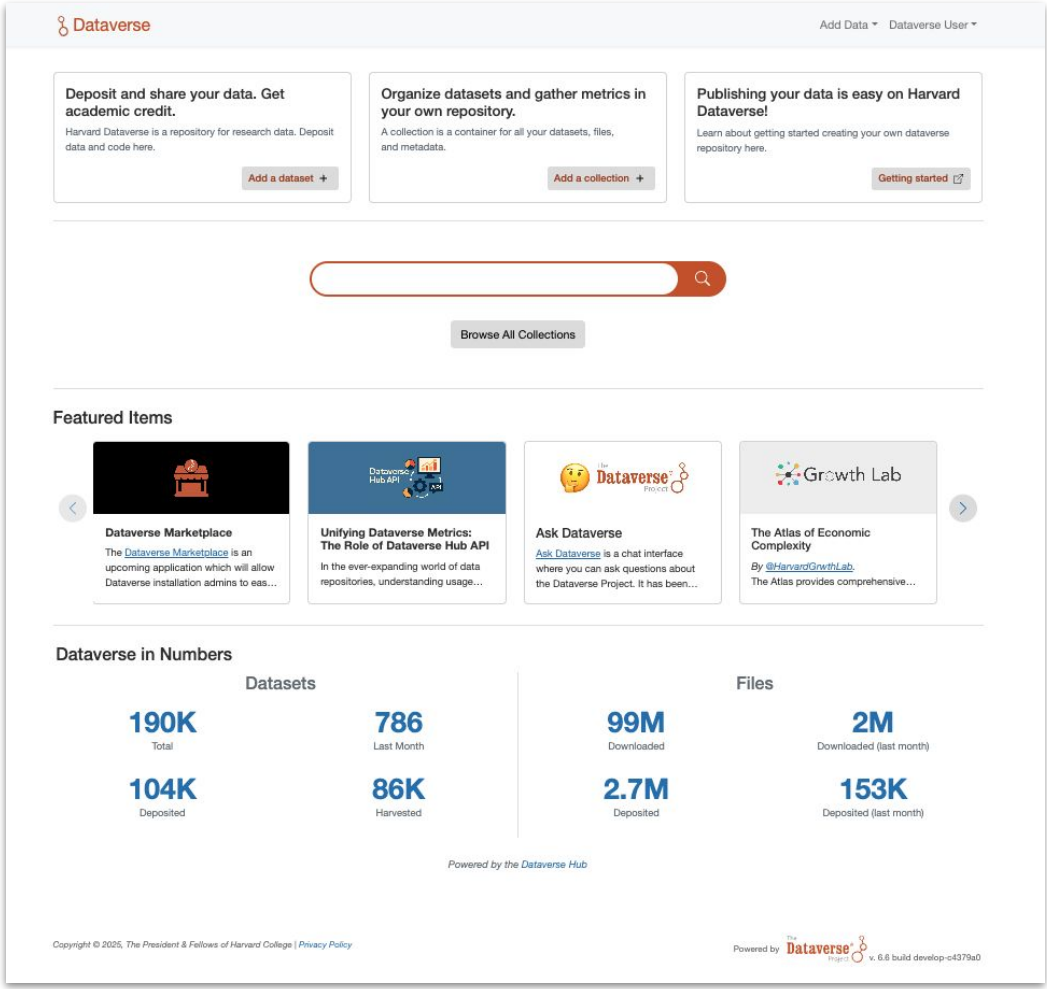
Design System & Mobile Responsiveness

Design System @iqss/dataverse-design-system

Reusable and modular design system:

- Ensures UI consistency across pages and features
- Promotes developer efficiency by using predefined components
- Provides a modern, accessible user interface
- Mobile Responsiveness layouts adjust to screen width


Dataverse
Browse All Collections
Deposit and share your data. Get academic credit.
Harvard Dataverse is a repository for research data. Deposit data and code here.
Add a dataset +
Organize datasets and gather metrics in your own repository.
A collection is a container for all your datasets, files, and metadata.
Add a collection +
Publishing your data is easy on Harvard Dataverse!
Learn about getting started creating your own dataverse repository here.
Getting started
Featured Items
Dataverse Marketplace
The Dataverse Marketplace is an upcoming application which will allow Dataverse installation admins to easi...
Unifying Dataverse Me
Role of Dataverse Hub
In the ever-expanding world
repositories, understanding
Dataverse in Numbers
Datasets
190K
Total
786
Last Month
104K
86K

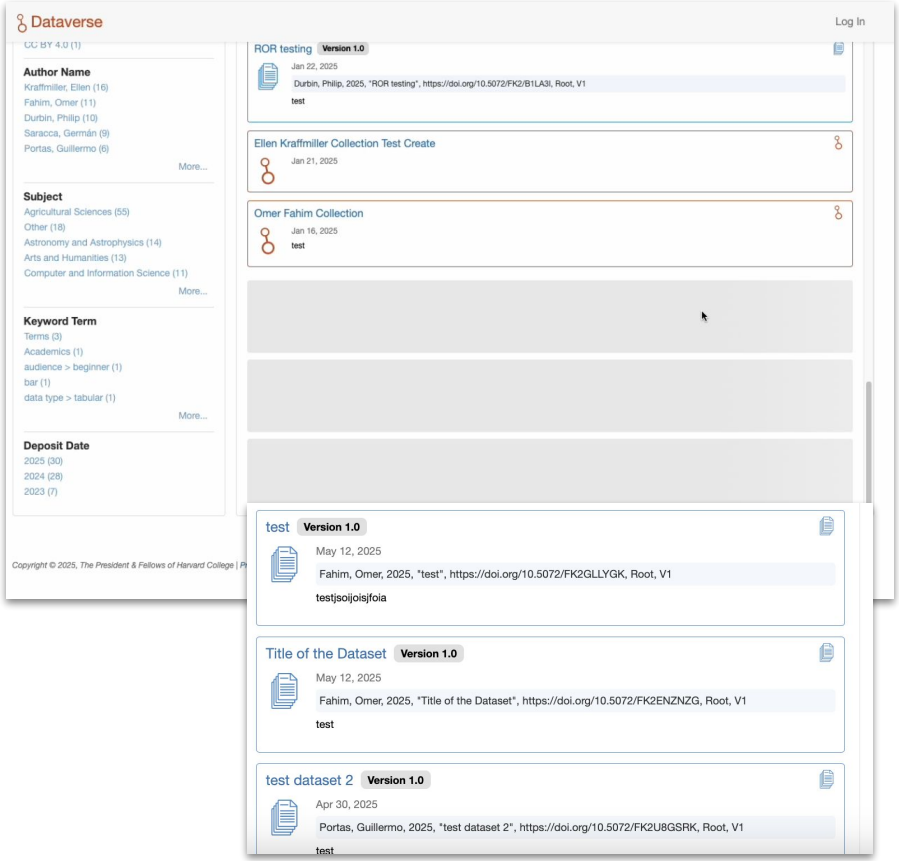

Dataverse
Add Data
Dataverse User
Deposit and share your data. Get academic credit.
Harvard Dataverse is a repository for research data. Deposit data and code here.
Add a dataset +
Organize datasets and gather metrics in your own repository.
A collection is a container for all your datasets, files, and metadata.
Add a collection +
Publishing your data is easy on Harvard Dataverse!
Learn about getting started creating your own dataverse repository here.
Getting started
Browse All Collections
Featured Items
Dataverse Marketplace
The Dataverse Marketplace is an upcoming application which will allow Dataverse installation admins to eas...
Unifying Dataverse Metrics: The Role of Dataverse Hub API
In the ever-expanding world of data repositories, understanding usage...
Ask Dataverse
Ask Dataverse is a chat interface where you can ask questions about the Dataverse Project. It has been...
Growth Lab
The Atlas of Economic Complexity
By @HarvardGrowthLab.
The Atlas provides comprehensive...
Dataverse in Numbers
Datasets
190K
Total
786
Last Month
104K
Deposited
86K
Harvested
Files
99M
Downloaded
2M
Downloaded (last month)
2.7M
Deposited
153K
Deposited (last month)
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Infinite Scrolling

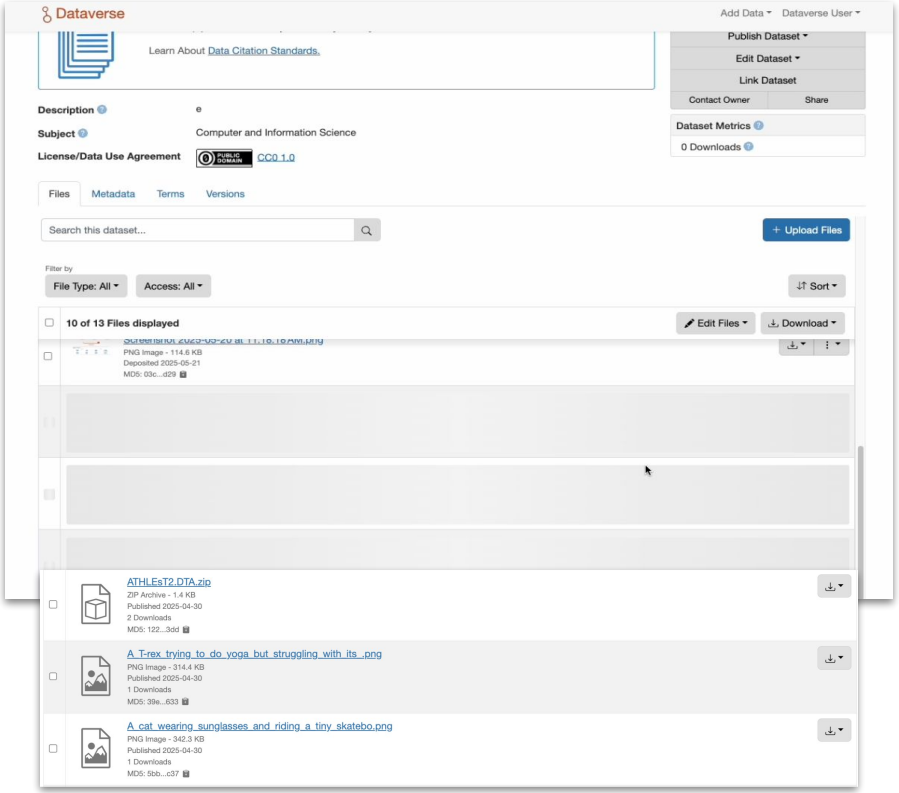
Infinite scrolling is a web design technique that loads more content dynamically as the user scrolls, eliminating the need for pagination.

- Provides a smooth, uninterrupted browsing experience.
- Enhances usability and engagement on smaller screens.

Collection Page

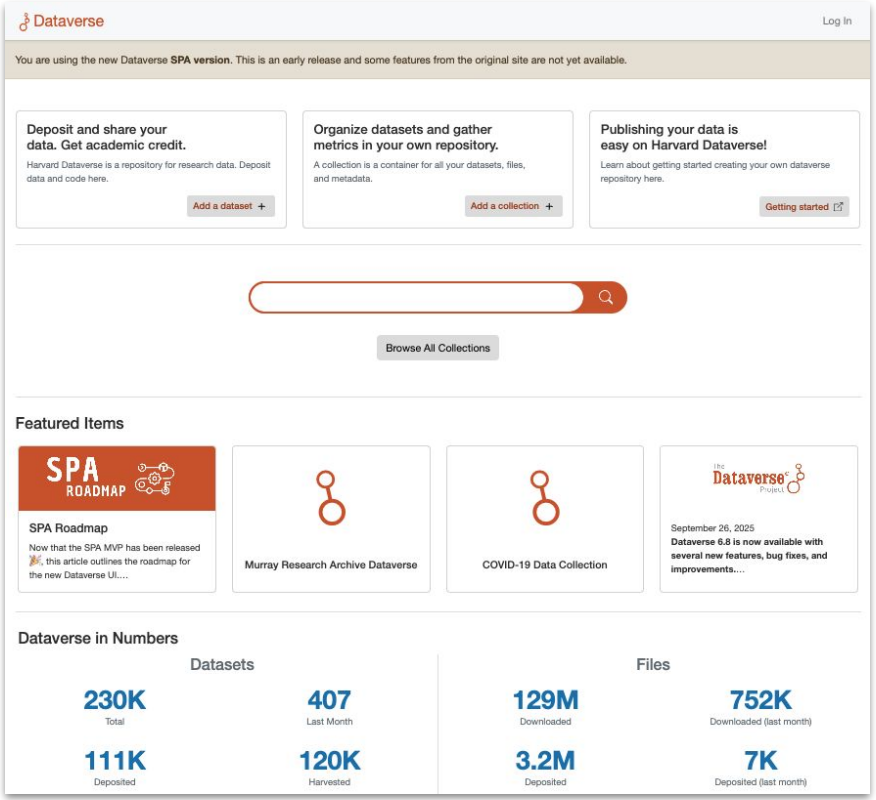


Dataset Page



⏰ Coming Soon: Official New UI Rollout

- Public soft launch later this year
- Over 60% of features available today
- We anticipate feature completeness by the end of the year



Thank you

Credits

- Stefano Iacus, Ellen Kraffmiller & Cheng Shi for Dataverse redesign slides
- Harvard Biomedical Data Management - RDM Lifecycle
- Flaticon: Image Stickers



Discussion

Resources

Title	Link
The Dataverse Project	https://dataverse.org/
The Harvard Dataverse Repository	https://dataverse.harvard.edu/
The repository support website	https://support.dataverse.harvard.edu/
Project Github	https://github.com/IQSS/dataverse
Support at Harvard Dataverse	support@dataverse.org
Presentations and publications	https://dataverse.org/presentations
NIH-GREI on Zenodo	https://datascience.nih.gov/data-ecosystem/generalist-repository-ecosystem-initiative